

Table 3: Effect of LLM exposure on firm-level job outcomes

	Employees	Low seniority	High seniority	Frac. low seniority	Av. comp.
LLM exposure x Post	-0.003* (0.002)	-0.004** (0.002)	-0.002 (0.002)	-0.001** (0.0003)	106.313* (59.029)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	1,268,013	1,267,955	1,266,331	1,267,955	1,268,013
Adj. R-squared	0.983	0.983	0.981	0.979	0.978

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, averaged over the post-treatment period. *Employees* is the total number of jobs in the firm. *Low seniority* and *High seniority* denote the number of positions in seniority levels 1–3 and 4–7, respectively. *Low seniority (%)* is the share of jobs that are low-seniority. *Av. comp.* is the mean annual salary plus bonuses. All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level and reported in parentheses. Significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The regression results in Table 3 show, averaged over the entire post-treatment period, a significant 0.4 percent reduction in junior employees per standard deviation increase in exposure, and a statistically insignificant reduction in senior employees. The fraction of junior employees goes down by 0.1 percentage points per standard deviation of LLM exposure ($p = 0.028$).

The observed compositional shift towards more senior roles has direct implications for compensation patterns. Figure 3 shows that average compensation rises modestly in highly exposed firms following the introduction of ChatGPT, with a one standard deviation increase in exposure associated with a £106 increase in average compensation, although the overall estimate is only marginally significant.

4.2 Job Posting Dynamics

4.2.1 Firm-Level Postings

While employment stocks adjust slowly, job postings provide a real-time window into firms’ evolving labor demand. We therefore examine how firms’ hiring intentions respond to LLM exposure by analyzing whether exposed firms post fewer vacancies overall and whether they adjust the composition of roles they seek to fill. This posting data captures the extensive margin of recruitment activity and reveals forward-looking decisions that precede the employment stock changes documented above. By tracking the same firms across both datasets, we can directly observe

Table 4: Effect of LLM exposure on occupation-level job postings

	Job postings (log)	Salary	Salary (log)	Salary (SD)
LLM exposure x Post	-0.039** (0.018)	-495.483*** (135.270)	-0.011*** (0.003)	-56.802 (108.358)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,815	16,815	16,815	16,804
Adjusted R ²	0.962	0.948	0.953	0.729

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level outcomes. *Job postings (log)* is the log number of new vacancies in occupation o in month t . *Salary* is the mean advertised annual compensation, *Salary (log)* is its logarithm, and *Salary (SD)* is the within-occupation standard deviation of posted salaries. *Exposure* is the occupation-level LLM exposure score from Eloundou et al. (2024). *Post* equals one for November 2022 onward. Standard errors are clustered at the occupation level. Significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.2.2 Occupation-Level Postings

This analysis examines occupation-level job posting data to understand how aggregate labor demand evolves for specific roles. This analysis exploits variation in LLM exposure across 384 occupations, from minimally affected roles such as cooks (exposure score: 0) to highly exposed positions like proofreaders and copy markers (exposure score: 0.86). We examine three outcomes: posting volumes to measure aggregate demand, advertised salaries to assess compensation responses, and within-occupation salary dispersion to measure wage polarization.

Figure 6 presents event study estimates for job posting volumes. Prior to November 2022, high- and low-exposure occupations follow parallel trajectories, validating our identification strategy. Following ChatGPT’s release, however, exposed occupations experience a sustained reduction in hiring demand, with effects first becoming statistically significant after 7 months and intensifying thereafter. After 20 months, we see a slight resurgence, and job postings converge back to their original levels. When averaged across the post-treatment period (Table 4), a one standard deviation increase in LLM exposure is associated with a 3.9% reduction in posting volume ($p = 0.029$). This is an economically meaningful change.

Turning to compensation, we observe immediate and persistent wage pressure for exposed occupations. Figure 7 indicates that posted salaries begin declining shortly after ChatGPT’s introduction. A one standard deviation increase in LLM exposure is associated with a £495 decline in posted salaries and a 1.1% reduction in log

quartile serves as the reference category. For firm-level job postings, effects are significant across all higher quartiles, though the third and fourth quartiles show similar magnitudes, which is consistent with a threshold effect. For firm-level employment stocks, the evidence is weaker and non-monotonic, with only scattered significant effects across bins. The occupation-level analysis reveals clearer nonlinearities: while the second and third quartiles show minimal effects, the highest exposure quartile experiences significantly larger impacts, suggesting that most of the occupation-level results we observe are driven by the 25 percent most exposed occupations.

5 Conclusion

This paper provides evidence that large language models are reshaping UK labor markets. Using a difference-in-differences design that exploits variation in occupational- and firm-level LLM exposure, we conduct three analyses that show how labor markets change in response to the introduction of generative AI. First, we examine firm-level employment stocks to understand how workforce size and composition evolve within organizations. Second, we analyze firm-level hiring patterns to capture forward-looking recruitment decisions that precede employment adjustments. Third, we investigate occupation-level posting dynamics to identify economy-wide demand shifts.

Our firm-level employment analysis documents a 0.3 percent reduction in total employment per standard deviation increase in LLM exposure, though this effect only becomes statistically significant approximately 18 months after ChatGPT’s introduction. This reduction is predominantly concentrated in junior positions, which decline by 0.4 percent per standard deviation, while senior positions show a smaller, statistically insignificant 0.2 percent reduction. The resulting compositional shift reduces the fraction of junior workers by 0.1 percentage point per standard deviation, mechanically increasing average compensation by £106 as firms retain proportionally more higher-paid senior staff, though this compensation effect is only marginally significant.

While employment stocks adjust gradually, job posting data provide real-time evidence of firms’ evolving labor demand. Firm-level posting activity drops following

ChatGPT’s release, with a one standard deviation increase in LLM exposure reducing the probability of posting any vacancy by 1.0 percentage point. This reduction emerges gradually and becomes significant after approximately 10 months, mirroring the timeline of employment stock adjustments. When we examine the composition of roles driving this reduction, we find that technical positions and creative roles experience the sharpest declines. Sales representative positions, by contrast, show a statistically significant increase, suggesting that customer-facing roles requiring direct interpersonal interaction remain robust to LLM-based automation.

In addition to within-firm dynamics, our occupation-level analysis reveals how labor demand shifts for entire job categories regardless of firm exposure. A one standard deviation increase in LLM exposure is associated with a 3.9 percent reduction in posting volumes, accompanied by £495 (1.1 percent) salary declines. The apparent inconsistency of declining posted salaries for exposed occupations alongside increasing firm-level compensation shows how aggregation could mask opposing forces. While LLM exposure drives down market wages for specific occupations, within-firm rebalancing toward more senior positions pushes average firm-level wages in the opposite direction.

Our heterogeneity analysis reveals that the observed labor market disruptions are almost entirely driven by high-remuneration segments of the labor market. This concentration of effects in higher-wage segments contradicts historical automation patterns that predominantly displaced routine middle-skill work, and suggests that LLMs might lead to a compression of the wage distribution.

Our findings carry important implications for understanding technological displacement and designing policy responses. The concentration of job losses among entry-level positions disrupts traditional skill development pathways where workers master progressively complex tasks through hands-on experience. Without junior roles serving as training grounds, firms may struggle to develop senior talent internally while new entrants face shrinking opportunities to gain experience (Ide, 2025). The freeze in hiring followed by employment stock changes suggests that forward-looking indicators like job postings may provide early warning signals for policymakers. The sharp reductions in technical and analytical roles alongside resilience in sales positions indicates that human interaction and relationship-building skills remain valuable complements to LLM capabilities, suggesting that retraining programs should emphasize these interpersonal competencies. Policy responses